Yassine Chouk

Embedded Systems Engineering Student | Seeking an Embedded Firmware Internship

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Summary

Embedded systems engineering student at INSAT building real-time firmware in embedded C/C++ on STM32, ESP32-S3 and TI C2000, using FreeRTOS and bare-metal programming. Experience in motor control (PWM), a dual-slot OTA bootloader, UART communication and hardware-in-the-loop (HIL) testing.

Education

Engineering Degree – Industrial IT & Automation Sep 2023 – Jun 2028 (expected)
National Institute of Applied Science and Technology (INSAT), Tunis

Baccalauréat – Experimental Sciences, Honors (17.44/20), Lycée Taher Sfar, Mahdia Jun 2023

Projects

FreeRTOS Inspector – Live Timing Analyzer on a HIL Motor Bench Oct 2026
STM32L476RG, FreeRTOS, ESP32-S3, DWT, UART | github.com/yassinechouk/Freertos-inspector

- Built a non-intrusive FreeRTOS analyzer that auto-discovers tasks and measures response time and deadline misses, with a live web dashboard; 1.7 % CPU overhead, 4 CRC errors in 51 886 frames
- Used response-time analysis to predict deadline misses (7 561 before), then re-assigned priorities: 0 misses on 13 tasks at 47.8 % CPU; validated on a motor bench (103 ms rise time matching simulation) and a fault lab that detected 5 of 5 injected faults within 0.58 s

STM32 Dual-Slot OTA Bootloader Sep 2026
STM32L476RG, Bare-Metal C, ESP32-S3, UART, CRC32, IWDG, Python | github.com/yassinechouk/BOOTLOADER_OTA

- Wrote a bare-metal dual-slot bootloader from scratch: three-layer verification (per-frame CRC32, write read-back, whole-image CRC), power-loss-safe duplicated metadata, and automatic rollback by the independent watchdog after 3 failed boots
- Built a framed UART update protocol, an ESP32 Wi-Fi gateway and Python host tools for wireless updates; validated with a 43-test host suite

Trackify – Embedded Safety Device for Alzheimer's Patients Dec 2025
ESP32-S3, MPU6050, SIM800L, NEO-6M GPS | github.com/yassinechouk/smart-belt-alzheimer

- Developed a fall-detection device running a Random Forest (trained on SisFall) on the ESP32-S3 to classify idle, walking and falling in real time, with Wi-Fi cloud alerts and GSM SMS fallback

AQUA GENIUS – Smart Autonomous Irrigation System Oct 2025
ESP32-S3, soil moisture and DS18B20 sensors, relay, XGBoost | github.com/yassinechouk/AQUA_GENIUS_project

- Trained XGBoost models for pump control and water volume and ported them to C++ on the ESP32-S3: under 10 ms inference, about 400 KB flash, about 1 KB RAM, bit-exact match with the Python output

Professional Experience

Embedded Control Engineering Intern – Qehna LAB, ENIT, Tunis Jun 2026 (1 month)
Power Quality with Power Electronics & Advanced Control | github.com/yassinechouk/c2000-vf-motor-control

- Implemented open-loop scalar V/f control of a three-phase induction motor on a TI F28379D (C2000): three-zone boost/linear/saturation V/f law and third-harmonic injection PWM (about 15.5 % more DC-bus utilisation)
- Generated 120-degree phase-shifted sine references with the TMU hardware instruction and verified the 10 kHz carrier and 300 ns dead-time on an oscilloscope

Test Automation Intern, End-of-Line Test – Sagemcom, Ezzahra Aug 2025 (1 month)
Production test bench log analysis | github.com/yassinechouk/LOGCOMPARE

- Built LOGCOMPARE, a Python/Tkinter tool that compares test bench logs against a reference with colour-coded thresholds and exports to Excel with charts; replaced an estimated 30 min of manual inspection per cycle

Skills

Programming Languages: C, C++, Python
Platforms: STM32, ESP32-S3, TI C2000, Raspberry Pi
RTOS & Firmware: FreeRTOS, Real-Time Scheduling, Bare-Metal, Bootloader, OTA Updates, Interrupts, DMA, ADC, DAC, Timers, PWM
Protocols: UART, SPI, I2C, CAN, Wi-Fi
Tools: STM32CubeIDE, Code Composer Studio, ESP-IDF, Arduino IDE, Git, Linux
Testing: Hardware-in-the-Loop (HIL), On-Target Debugging, Oscilloscope, Logic Analyzer

Languages

Arabic (Native) | English (IELTS 7.0, C1) | French (C1, Advanced) | German (ÖSD A2)